

Quiz: Multiple Integrals

Companion quiz to “Lecture Notes: Multiple Integrals” (covers Sections 1–6: iterated integrals, general regions, changing order, polar coordinates, and triple integrals).

A. Conceptual / Visualization

A1. Sketch the region D bounded by $y = x^2$ and $y = \sqrt{x}$ for $0 \leq x \leq 1$. Shade it, and mark the two points where the curves intersect.

A2. Sketch the triangular region with vertices $(0, 0)$, $(0, 1)$, $(1, 1)$ that appears in $\int_0^1 \int_x^1 e^{y^2} dy dx$.

Then rewrite the integral in the order of $dx dy$.

A3. Without doing any computation, describe in words what the solid $E = \{(x, y, z) : x^2 + y^2 \leq 4, 0 \leq z \leq 4 - x^2 - y^2\}$ looks like, and state which coordinate system (rectangular, polar-in- xy , or otherwise) is most natural for projecting it onto the xy -plane.

A4. True or false, and justify briefly: “For triple integrals, any of the six possible orders of integration ($dx dy dz$, $dz dx dy$, etc.) are equally natural to set up for a general solid E .” If false, explain which variable is usually best kept outermost and why.

A4. True or false, “polar coordinates can be used in triple integrals”.

B. Error Analysis (Spot the Mistake)

Each item below contains a setup error. Identify the error(s) and correct them.

B1. A student converts $\iint_D (x^2 + y^2) dA$, where D is the disk $x^2 + y^2 \leq 9$, to polar coordinates and writes:

$$\iint_D (x^2 + y^2) dA = \int_0^{2\pi} \int_0^3 r^2 dr d\theta$$

B2. A student sets up a Type I double integral as:

$$\int_0^{g_1(x)} \int_a^b f(x, y) dy dx$$

B3. A student is asked to switch the order of integration on $\int_0^1 \int_x^1 e^{y^2} dy dx$ and simply writes:

$$\int_0^1 \int_x^1 e^{y^2} dy dx = \int_x^1 \int_0^1 e^{y^2} dx dy$$

B4. A student sets up a triple integral over a solid E as:

$$\iiint_E f dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x,y,z)}^{u_2(x,y,z)} f dz dy dx$$

Explain why this is impossible as written, independent of what f or E actually are.

C. Set Up Only (Do Not Evaluate)

C1. Set up the double integral $\iint_D xy \, dA$, where D is the region bounded by $y = x$ and $y = x^3$ in the first quadrant.

C2. Set up the integral needed to reverse the order of integration in $\int_0^1 \int_{y^2}^1 f(x, y) \, dx \, dy$ (i.e., rewrite it as $dy \, dx$ instead). Do not evaluate.

C3. Set up, in polar coordinates, the double integral for the area that lies inside both circles $r = 1$ and $r = 2 \sin \theta$.

C4. Set up the triple integral, with z outermost, for the volume of the solid bounded above by $z = x + y + 2$ and below by the rectangle $D = [0, 1] \times [0, 1]$ in the xy -plane.

D. Short Computation

D1. Evaluate $\int_0^1 \int_0^2 (3x + 2y) \, dy \, dx$.

D2. Evaluate $\int_0^1 \int_0^x 6xy \, dy \, dx$.

D3. Use polar coordinates to evaluate $\iint_D 1 \, dA$, where D is the quarter-disk $x^2 + y^2 \leq 1$, $x \geq 0$, $y \geq 0$ (i.e., confirm this equals the expected quarter-circle area).

D4. Evaluate $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} dz \, dy \, dx$ (the volume of the tetrahedron bounded by the coordinate planes and the plane $x + y + z = 1$).

Answer Key (Brief)

A1: The two curves meet where $x^2 = \sqrt{x}$, i.e., at $x = 0$ and $x = 1$; on $(0, 1)$, $\sqrt{x} \geq x^2$, so D is the lens-shaped region between $y = x^2$ (bottom) and $y = \sqrt{x}$ (top).

A2: As Type I, the triangle is described by $0 \leq x \leq 1$, $x \leq y \leq 1$ (vertical strips between the line $y = x$ and the line $y = 1$). As Type II, the same triangle is $0 \leq y \leq 1$, $0 \leq x \leq y$ (horizontal strips between $x = 0$ and the line $x = y$) — same region, different slicing direction.

A3: E is the solid sitting above the disk $x^2 + y^2 \leq 4$ in the xy -plane and below the downward paraboloid $z = 4 - x^2 - y^2$ — a dome-shaped solid. Since the projection D is a disk and the integrand/boundary involves $x^2 + y^2$, polar coordinates in x, y (with z kept rectangular) is the natural choice for the outer double integral.

A4: False. It's best practice to keep z outermost (i.e., use $dz \, dy \, dx$ or $dz \, dx \, dy$) because the solid is naturally trapped between a bottom and top surface in z , which gives simple z -bounds; the remaining double integral over the xy -projection can then be handled with familiar Type I/Type II/polar techniques. Making x or y outermost instead usually forces an awkward, unnatural description of the solid.

B1: Missing the extra factor of r from the polar area element $dA = r \, dr \, d\theta$, and the integrand $x^2 + y^2 = r^2$ needs that extra r folded in too. Correct: $\int_0^{2\pi} \int_0^3 r^2 \cdot r \, dr \, d\theta = \int_0^{2\pi} \int_0^3 r^3 \, dr \, d\theta$.

B2: The inner variable's bounds (0 to $g_1(x)$) don't match a Type I region correctly here — more importantly, the outer variable's bounds (a to b) are written for x , but x is the *inner* differential in this integral ($dy \, dx$ means y is inner, x is outer) — so the bounds are simply mismatched to the

wrong variable: y 's bounds should depend on x (e.g., $g_1(x)$ to $g_2(x)$), while x 's bounds should be constants (a to b), not the other way around.

B3: Simply swapping $dy dx \rightarrow dx dy$ without changing the limits is incorrect — the bounds x to 1 and 0 to 1 don't describe the same region once the order is swapped. The correct re-derivation (see Section 4, Worked Example 1) is $\int_0^1 \int_0^y e^{y^2} dx dy$, found by re-sketching the triangular region and re-describing it as Type II.

B4: This violates the general bounds rule: the innermost variable being integrated (z) cannot appear in its own bounds. The u_1, u_2 bounds for z are only allowed to depend on the *outer* variables not yet integrated (here, x and y), never on z itself.

$$\mathbf{D1:} \quad \int_0^1 \int_0^2 (3x + 2y) dy dx = \int_0^1 [3xy + y^2]_0^2 dx = \int_0^1 (6x + 4) dx = 3 + 4 = 7.$$

$$\mathbf{D2:} \quad \int_0^1 \int_0^x 6xy dy dx = \int_0^1 3x \cdot x^2 dx = \int_0^1 3x^3 dx = \frac{3}{4}.$$

$$\mathbf{D3:} \quad \int_0^{\pi/2} \int_0^1 r dr d\theta = \frac{\pi}{2} \cdot \frac{1}{2} = \frac{\pi}{4}, \text{ matching the known quarter-circle area } \frac{1}{4}\pi(1)^2. \quad \checkmark$$

$$\mathbf{D4:} \quad \int_0^1 \int_0^{1-x} \int_0^{1-x-y} dz dy dx = \int_0^1 \int_0^{1-x} (1-x-y) dy dx = \int_0^1 \frac{(1-x)^2}{2} dx = \left[-\frac{(1-x)^3}{6} \right]_0^1 =$$

$$\frac{1}{6}.$$